1. What is the primary purpose of a hardened computer in industrial automation?

A) To run video games

B) To withstand harsh industrial environments

C) To replace HMIs

D) To act as a backup power supply

Answer: B

2. Which of these is NOT a component of a typical PLC setup?

A) Power Supply

B) I/O Modules

C) Cooling Fan

D) CPU

Answer: C

3. In PLC terminology, what does CHI stand for?

A) Computer Human Interface

B) Central Hardware Interconnect

C) Control Hub Integration

D) Closed Hardware Interface

Answer: A

4. Which PLC architecture allows for flexible expansion of I/O modules?

A) Fixed

B) Modular

C) Closed

D) Analog

Answer: B

5. What is the primary role of the right vertical line in traditional ladder logic diagrams?

A) Power conductor

B) Return path (neutral)

C) Ground connection

D) It is omitted in PLC ladder logic

Answer: D

6. Which of these tasks is automation LEAST likely to replace humans in?

A) Nuclear facility monitoring

B) Handling extremely heavy loads

C) Creative problem-solving

D) Repetitive assembly line work

Answer: C

7. What communication method is NOT typically used between a PLC and a programming device?

A) Ethernet

B) USB

C) Serial/parallel link

D) Wi-Fi

Answer: D

8. Which part of the PLC physically connects to field devices like sensors and actuators?

A) CPU

B) Power Supply

C) I/O Modules

D) Programming Device

Answer: C

9. What is a rung in ladder logic programming?

A) A single vertical power line

B) A horizontal line with control elements and one output coil

C) A type of CPU instruction

D) A modular I/O unit

Answer: B

10. Which IEC 61131 language uses text-based syntax similar to Pascal?

A) Ladder Logic (LD)

B) Structured Text (ST)

C) Function Block Diagram (FBD)

D) Sequential Function Chart (SFC)

Answer: B

11. What does PLC stand for?

A) Programmable Linear Controller

B) Programmable Logic Controller

C) Process Logic Computer

D) Power Line Communication

Answer: B

12. Which component is considered the "brain" of a PLC?

A) Power Supply

B) I/O Module

C) CPU

D) Programming Device

Answer: C

13. What is the primary function of the I/O module in a PLC?

A) To supply power to the CPU

B) To act as a bridge between field devices and the PLC

C) To program the ladder logic

D) To replace human operators

Answer: B

14. Which of the following is NOT a common PLC programming language under IEC 61131?

A) Ladder Logic (LD)

B) Python

C) Structured Text (ST)

D) Function Block Diagram (FBD)

Answer: B

15. What voltage range typically powers a PLC's internal logic?

A) 110-220V AC

B) 5-24V DC

C) 48V AC

D) 12-36V AC

Answer: B

16. What is the purpose of the scan cycle in a PLC?

A) To turn off the system

B) To execute the program repeatedly in real-time

C) To replace the CPU

D) To connect to the internet

Answer: B

17. Which of the following is an advantage of industrial automation?

A) Higher unpredictability

B) Replacing humans in dangerous environments

C) Increased manual labor costs

D) Slower production speeds

Answer: B

18. What does HMI stand for in industrial automation?

A) Human-Machine Interface

B) High-Memory Integration

C) Hardware Monitoring Interface

D) Hybrid Motor Input

Answer: A

19. Which PLC part converts AC voltage to DC for internal use?

A) CPU

B) Power Supply

C) I/O Module

D) Programming Device

Answer: B

20. In ladder logic, what does the left vertical line represent?

A) Neutral path

B) Power conductor

C) Output coil

D) Ground connection

Answer: B

21. What is the main disadvantage of automation?

A) Improved economy

B) High initial costs

C) Reduced human intervention

D) Faster task completion

Answer: B

22. Which PLC architecture can refer to hardware, software, or both?

A) Open or Closed

B) Fixed or Modular

C) Analog or Digital

D) Serial or Parallel

Answer: A

23. What type of signals do input modules typically receive?

A) Only digital

B) Only analog

C) Both digital and analog

D) Only wireless

Answer: C

24. Which device is most commonly used to program a PLC?

A) Smartphone

B) Oscilloscope

C) Personal Computer (PC)

D) Multimeter

Answer: C

25. What is the correct reading direction for ladder logic diagrams?

A) Right-to-left, bottom-to-top

B) Left-to-right, top-to-bottom

C) Top-to-bottom only

D) Random order

Answer: B

26. Which of the following is NOT a field device connected to PLC I/O modules?

A) Sensors

B) Actuators

C) USB Drives

D) Solenoids

Answer: C

27. What is the role of a stationery engineer in automation?

A) To repair PLC hardware

B) To monitor and control HMIs

C) To design new PLCs

D) To replace CPUs

Answer: B

28. Which PLC language resembles electrical relay logic diagrams?

A) Structured Text (ST)

B) Ladder Logic (LD)

C) Sequential Function Chart (SFC)

D) Instruction List (IL)

Answer: B

29. What does IEC 61131 standardize?

A) PLC hardware dimensions

B) PLC programming languages

C) Power supply voltages

D) HMI colors

Answer: B

30. Which PLC component stores user programs and I/O status?

A) Power Supply

B) Memory (in CPU)

C) I/O Module

D) Ethernet Cable

Answer: B

31. What is the key difference between fixed and modular I/O systems?

A) Modular systems are non-expandable

B) Fixed systems are cheaper

C) Modular systems allow customization

D) Fixed systems use analog signals only

Answer: C

32. How are ladder logic rungs read?

A) Right-to-left, bottom-to-top

B) Left-to-right, top-to-bottom

C) Top-to-bottom only

D) Randomly

Answer: B

33. Which element is ALWAYS present in a ladder logic rung?

A) Timer

B) Output coil

C) Counter

D) Analog input

Answer: B

34. What is the purpose of the "rack" in a PLC system?

A) To house expansion modules

B) To cool the CPU

C) To store programming software

D) To connect to Wi-Fi

Answer: A

35. Which programming language uses graphical blocks for logic?

A) Structured Text (ST)

B) Function Block Diagram (FBD)

C) Instruction List (IL)

D) Sequential Function Chart (SFC)

Answer: B

36. What is the primary drawback of high initial automation costs?

A) Limited scalability

B) Unpredictable ROI

C) Excessive energy use

D) Frequent hardware failures

Answer: B

37. Which PLC component directly interfaces with a motor actuator?

A) CPU

B) Output Module

C) Power Supply

D) Programming Device

Answer: B

38. What is the primary input to a PLC from a push button?

A) Analog signal

B) Digital signal

C) Wireless signal

D) Mechanical force

Answer: B

39. Which IEC 61131 language is sequential and state-based?

A) Ladder Logic (LD)

B) Sequential Function Chart (SFC)

C) Structured Text (ST)

D) Instruction List (IL)

Answer: B

40. Why are PLCs preferred over general-purpose computers in industry?

A) Lower cost

B) Real-time reliability

C) Better graphics

D) Internet connectivity

Answer: B

41. What is the critical function of the CPU's memory?

A) Store HMI graphics

B) Hold user programs and I/O status

C) Power field devices

D) Generate analog signals

Answer: B

42. What is the primary function of a relay in an electrical system?

A) To generate electrical power

B) To regulate the movement of mechanical contacts using an electromagnet

C) To store electrical energy

D) To measure voltage fluctuations

Answer: B

43. How does a contactor differ from a standard relay?

A) It uses a transistor instead of a triac

B) It is designed for high-power applications with larger contacts and coils

C) It operates only on direct current (DC)

D) It has no moving parts

Answer: B

44. Which of the following is NOT a type of relay construction?

A) Clapper type

B) Bridge type

C) Pneumatic type

D) Laminated type

Answer: D

45. What is the purpose of a shading coil in an AC relay?

A) To increase voltage output

B) To prevent contact chattering

C) To store electrical charge

D) To reduce electromagnetic interference

Answer: B

46. Which type of relay uses a semiconductor for switching without mechanical contacts?

A) Electromechanical relay

B) Thermal relay

C) Solid-state relay (SSR)

D) Pneumatic relay

Answer: C

47. What component does a solid-state relay use to control AC loads?

A) Transistor

B) Triac

C) Capacitor

D) Diode

Answer: B

48. Which timing relay classification is also known as "Delay On Energize" (DOE)?

A) Off-delay relay

B) On-delay relay

C) Interval timer

D) Capacitor limit timer

Answer: B

49. What is the main advantage of pneumatic timers in industrial applications?

A) They are unaffected by ambient temperature variations

B) They use digital displays for precise timing

C) They operate using microcontrollers

D) They require no air supply

Answer: A

50. Which type of timer uses a synchronous motor similar to a wall clock?

A) Electronic timer

B) Clock timer

C) Capacitor limit timer

D) Motor-driven timer

Answer: B

51. What is the primary function of an overload relay in a motor control system?

A) To regulate motor speed

B) To protect the motor from excessive current

C) To generate electrical signals

D) To measure voltage levels

Answer: B

52. Which type of overload relay operates based on heat generated by motor current?

A) Magnetic overload relay

B) Electronic overload relay

C) Thermal overload relay

D) Pneumatic overload relay

Answer: C

53. What is the key difference between a thermal and magnetic overload relay?

A) Thermal relays use heat, while magnetic relays use electromagnetic fields

B) Thermal relays are only for DC circuits

C) Magnetic relays are used in low-power applications

D) Thermal relays have no moving parts

Answer: A

54. Which device detects phase loss or imbalance in a three-phase system?

A) Contactor

B) Phase failure relay

C) Solid-state relay

D) Timing relay

Answer: B

55. What is the primary role of a contactor in an electrical system?

A) To provide overload protection

B) To switch high-power circuits electromagnetically

C) To measure current flow

D) To store electrical energy

Answer: B

56. Which of the following is NOT a method of isolation in solid-state relays?

A) Optical isolation

B) Transformer isolation

C) Reed isolation

D) Pneumatic isolation

Answer: D

57. What type of relay combines an electromechanical output relay with a control circuit for timed operations?

A) Thermal relay

B) Timing relay

C) Phase failure relay

D) Magnetic relay

Answer: B

58. Which of the following is a characteristic of electronic timers?

A) They use compressed air for timing

B) They are programmable with digital displays

C) They rely on mechanical gears only

D) They cannot be adjusted

Answer: B

59. What is the purpose of a triac in a solid-state relay?

A) To store electrical charge

B) To allow bidirectional current flow for AC loads

C) To generate electromagnetic fields

D) To measure temperature

Answer: B

60. Which of the following is true about pneumatic timers?

A) They use microcontrollers for timing

B) They are sensitive to electrical noise

C) They operate using compressed air

D) They cannot be adjusted

Answer: C

61. What is the function of a capacitor in a capacitor limit timer relay?

A) To store and release energy for timing control

B) To generate electromagnetic fields

C) To measure motor speed

D) To provide overload protection

Answer: A

62. Which type of relay is commonly used in PLCs for noise isolation?

A) Electromechanical relay

B) Thermal relay

C) Optoisolated solid-state relay

D) Pneumatic relay

Answer: C

63. What happens when an overload relay trips?

A) The motor speed increases

B) The circuit is interrupted to protect the motor

C) The voltage level rises

D) The relay emits light

Answer: B

64. Which of the following is a disadvantage of thermal overload relays?

A) They are unaffected by ambient temperature

B) They trip faster in warm environments

C) They cannot be used with motors

D) They require digital displays

Answer: B

65. What is the main advantage of solid-state relays over electromechanical relays?

A) They have moving contacts that wear out

B) They are slower in operation

C) They provide electrical noise isolation

D) They cannot handle high power

Answer: C

66. Which symbol represents an overload relay in schematic diagrams?

A) A circle with a "T" inside

B) A rectangle with diagonal lines

C) A zigzag line inside a square

D) A coil symbol with contacts

Answer: B

67. What is the primary purpose of a clamp-on ammeter in troubleshooting?

A) To measure voltage drop across a component

B) To measure current without breaking the circuit

C) To test continuity of a de-energized circuit

D) To generate a short-circuit signal

Answer: B

68. In a live circuit, if a voltmeter reads 0V across a closed switch, what does this indicate?

A) The switch is defective (open)

B) The switch is functioning correctly (closed)

C) The power supply is disconnected

D) The voltmeter is faulty

Answer: B

69. When using an ohmmeter to test a relay coil, what must be ensured first?

A) The circuit is energized

B) The power is disconnected

C) The relay is in motion

D) The meter is set to measure voltage

Answer: B

70. What is the purpose of a fused jumper in troubleshooting?

A) To permanently bypass a component

B) To safely simulate a closed switch or contact during testing

C) To measure resistance in a live circuit

D) To increase current flow to the load

Answer: B

71. When troubleshooting a motor overload, which instrument would confirm excessive current draw?

A) Ohmmeter

B) Clamp-on ammeter

C) Voltmeter

D) Megohmmeter

Answer: B

72. What critical safety step must be taken before using an ohmmeter to test a control circuit?

A) Energize the circuit to check for voltage

B) Disconnect power and verify isolation

C) Ground the meter leads

D) Set the meter to measure AC current

Answer: B

73. In a ladder diagram, what do the two vertical "rails" represent?

A) Physical location of components

B) Live and neutral power lines

C) Wire colors in the circuit

D) Mechanical linkages

Answer: B

74. How are STOP-function devices typically connected in a ladder diagram?

A) In parallel with the load

B) In series between the left rail and load

C) From the left rail directly to the right rail

D) Inside the load itself

Answer: B

75. Which type of limit switch uses a roller for actuation?

A) Plunger switch

B) Spring arm switch

C) Roller lever switch

D) Proximity switch

Answer: C

76. What does a flow switch monitor?

A) Electrical resistance

B) Fluid flow in a system

C) Light intensity

D) Magnetic fields

Answer: B

77. What is another name for an air flow switch?

A) Sail switch

B) Thermal switch

C) Float switch

D) Selector switch

Answer: A

78. What is the function of a temperature switch (thermostat)?

A) To measure electrical current

B) To monitor and control temperature thresholds

C) To generate hydraulic pressure

D) To store thermal energy

Answer: B

79. How is a selector switch operated?

A) By pressing a button

B) By rotating a knob

C) By voice command

D) By detecting motion

Answer: B

80. What is a common configuration for a selector switch in industrial controls?

A) NO-NC-ON

B) MAN-OFF-AUTO

C) HIGH-LOW-OFF

D) START-STOP-RESET

Answer: B

81. What is the main difference between a push button and a selector switch?

A) Push buttons use light, selector switches use sound

B) Push buttons are pressed, selector switches are rotated

C) Push buttons are electronic, selector switches are mechanical

D) Push buttons have no contacts

Answer: B

82. Which switch is used to detect the absence of fluid flow in a system?

A) Limit switch

B) Flow switch

C) Pressure switch

D) Temperature switch

Answer: B

83. What happens when a flow switch detects no fluid flow?

A) It emits light

B) It triggers an alarm or stops a pump

C) It increases pressure

D) It locks in place

Answer: B

84. Which type of switch is commonly used in HVAC systems?

A) Float switch

B) Temperature switch

C) Selector switch

D) Push-pull button

Answer: B

85. What is the purpose of the "bumper arm" in a limit switch?

A) To store energy

B) To interact with an object (e.g., cam)

C) To measure voltage

D) To generate heat

Answer: B

86. Which switch type is mechanically operated by liquid levels?

A) Pressure switch

B) Float switch

C) Limit switch

D) Selector switch

Answer: B

87. What is the key advantage of a MAN-OFF-AUTO selector switch?

A) It has no moving parts

B) It provides manual, off, and automatic control modes

C) It operates only in high-voltage systems

D) It cannot be adjusted

Answer: B

88. What is the primary function of a relay in an electrical system?

A) To generate electrical power

B) To regulate the movement of mechanical contacts using an electromagnet

C) To store electrical energy

D) To measure voltage fluctuations

Answer: B

89. How does a contactor differ from a standard relay?

A) It uses a transistor instead of a triac

B) It is designed for high-power applications with larger contacts and coils

C) It operates only on direct current (DC)

D) It has no moving parts

Answer: B

90. Which of the following is NOT a type of relay construction?

A) Clapper type

B) Bridge type

C) Pneumatic type

D) Laminated type

Answer: D

91. What is the purpose of a shading coil in an AC relay?

A) To increase voltage output

B) To prevent contact chattering

C) To store electrical charge

D) To reduce electromagnetic interference

Answer: B

92. Which type of relay uses a semiconductor for switching without mechanical contacts?

A) Electromechanical relay

B) Thermal relay

C) Solid-state relay (SSR)

D) Pneumatic relay

Answer: C

93. What component does a solid-state relay use to control AC loads?

A) Transistor

B) Triac

C) Capacitor

D) Diode

Answer: B

94. Which timing relay classification is also known as "Delay On Energize" (DOE)?

A) Off-delay relay

B) On-delay relay

C) Interval timer

D) Capacitor limit timer

Answer: B

95. What is the main advantage of pneumatic timers in industrial applications?

A) They are unaffected by ambient temperature variations

B) They use digital displays for precise timing

C) They operate using microcontrollers

D) They require no air supply

Answer: A

96. Which type of timer uses a synchronous motor similar to a wall clock?

A) Electronic timer

B) Clock timer

C) Capacitor limit timer

D) Motor-driven timer

Answer: B

97. What is the primary function of an overload relay in a motor control system?

A) To regulate motor speed

B) To protect the motor from excessive current

C) To generate electrical signals

D) To measure voltage levels

Answer: B

98. Which type of overload relay operates based on heat generated by motor current?

A) Magnetic overload relay

B) Electronic overload relay

C) Thermal overload relay

D) Pneumatic overload relay

Answer: C

99. What is the key difference between a thermal and magnetic overload relay?

A) Thermal relays use heat, while magnetic relays use electromagnetic fields

B) Thermal relays are only for DC circuits

C) Magnetic relays are used in low-power applications

D) Thermal relays have no moving parts

Answer: A

100. Which device detects phase loss or imbalance in a three-phase system?

A) Contactor

B) Phase failure relay

C) Solid-state relay

D) Timing relay

Answer: B

101. What is the primary role of a contactor in an electrical system?

A) To provide overload protection

B) To switch high-power circuits electromagnetically

C) To measure current flow

D) To store electrical energy

Answer: B

102. Which of the following is NOT a method of isolation in solid-state relays?

A) Optical isolation

B) Transformer isolation

C) Reed isolation

D) Pneumatic isolation

Answer: D

103. What type of relay combines an electromechanical output relay with a control circuit for timed operations?

A) Thermal relay

B) Timing relay

C) Phase failure relay

D) Magnetic relay

Answer: B

104. Which of the following is a characteristic of electronic timers?

A) They use compressed air for timing

B) They are programmable with digital displays

C) They rely on mechanical gears only

D) They cannot be adjusted

Answer: B

105. What is the purpose of a triac in a solid-state relay?

A) To store electrical charge

B) To allow bidirectional current flow for AC loads

C) To generate electromagnetic fields

D) To measure temperature

Answer: B

106. Which of the following is true about pneumatic timers?

A) They use microcontrollers for timing

B) They are sensitive to electrical noise

C) They operate using compressed air

D) They cannot be adjusted

Answer: C

107. What is the function of a capacitor in a capacitor limit timer relay?

A) To store and release energy for timing control

B) To generate electromagnetic fields

C) To measure motor speed

D) To provide overload protection

Answer: A

108. Which type of relay is commonly used in PLCs for noise isolation?

A) Electromechanical relay

B) Thermal relay

C) Optoisolated solid-state relay

D) Pneumatic relay

Answer: C

109. What happens when an overload relay trips?

A) The motor speed increases

B) The circuit is interrupted to protect the motor

C) The voltage level rises

D) The relay emits light

Answer: B

110. Which of the following is a disadvantage of thermal overload relays?

A) They are unaffected by ambient temperature

B) They trip faster in warm environments

C) They cannot be used with motors

D) They require digital displays

Answer: B

111. What is the main advantage of solid-state relays over electromechanical relays?

A) They have moving contacts that wear out

B) They are slower in operation

C) They provide electrical noise isolation

D) They cannot handle high power

Answer: C

112. Which symbol represents an overload relay in schematic diagrams?

A) A circle with a "T" inside

B) A rectangle with diagonal lines

C) A zigzag line inside a square

D) A coil symbol with contacts

Answer: B

113. What is the primary purpose of a clamp-on ammeter in troubleshooting?

A) To measure voltage drop across a component

B) To measure current without breaking the circuit

C) To test continuity of a de-energized circuit

D) To generate a short-circuit signal

Answer: B

114. In a live circuit, if a voltmeter reads 0V across a closed switch, what does this indicate?

A) The switch is defective (open)

B) The switch is functioning correctly (closed)

C) The power supply is disconnected

D) The voltmeter is faulty

Answer: B

115. When using an ohmmeter to test a relay coil, what must be ensured first?

A) The circuit is energized

B) The power is disconnected

C) The relay is in motion

D) The meter is set to measure voltage

Answer: B

116. What is the purpose of a fused jumper in troubleshooting?

A) To permanently bypass a component

B) To safely simulate a closed switch or contact during testing

C) To measure resistance in a live circuit

D) To increase current flow to the load

Answer: B

117. When troubleshooting a motor overload, which instrument would confirm excessive current draw?

A) Ohmmeter

B) Clamp-on ammeter

C) Voltmeter

D) Megohmmeter

Answer: B

118. What critical safety step must be taken before using an ohmmeter to test a control circuit?

A) Energize the circuit to check for voltage

B) Disconnect power and verify isolation

C) Ground the meter leads

D) Set the meter to measure AC current

Answer: B

119. In a ladder diagram, what do the two vertical "rails" represent?

A) Physical location of components

B) Live and neutral power lines

C) Wire colors in the circuit

D) Mechanical linkages

Answer: B

120. How are STOP-function devices typically connected in a ladder diagram?

A) In parallel with the load

B) In series between the left rail and load

C) From the left rail directly to the right rail

D) Inside the load itself

Answer: B

121. Which type of limit switch uses a roller for actuation?

A) Plunger switch

B) Spring arm switch

C) Roller lever switch

D) Proximity switch

Answer: C

122. What does a flow switch monitor?

A) Electrical resistance

B) Fluid flow in a system

C) Light intensity

D) Magnetic fields

Answer: B

123. What is another name for an air flow switch?

A) Sail switch

B) Thermal switch

C) Float switch

D) Selector switch

Answer: A

124. What is the function of a temperature switch (thermostat)?

A) To measure electrical current

B) To monitor and control temperature thresholds

C) To generate hydraulic pressure

D) To store thermal energy

Answer: B

125. How is a selector switch operated?

A) By pressing a button

B) By rotating a knob

C) By voice command

D) By detecting motion

Answer: B

126. What is a common configuration for a selector switch in industrial controls?

A) NO-NC-ON

B) MAN-OFF-AUTO

C) HIGH-LOW-OFF

D) START-STOP-RESET

Answer: B

127. What is the main difference between a push button and a selector switch?

A) Push buttons use light, selector switches use sound

B) Push buttons are pressed, selector switches are rotated

C) Push buttons are electronic, selector switches are mechanical

D) Push buttons have no contacts

Answer: B

128. Which switch is used to detect the absence of fluid flow in a system?

A) Limit switch

B) Flow switch

C) Pressure switch

D) Temperature switch

Answer: B

129. What happens when a flow switch detects no fluid flow?

A) It emits light

B) It triggers an alarm or stops a pump

C) It increases pressure

D) It locks in place

Answer: B

130. Which type of switch is commonly used in HVAC systems?

A) Float switch

B) Temperature switch

C) Selector switch

D) Push-pull button

Answer: B